

Lab Environment Preparation

Lab Objectives:

In this lab, you will prepare a complete penetration-testing environment using **VirtualBox**. By the end of this lab, you will have **Kali Linux** and **Metasploitable** running on the same isolated virtual network, able to communicate with each other and access the internet.

This environment will be used in upcoming labs.

Phase 1: Prerequisites & Dependencies

Before installing VirtualBox, ensure the following components are installed on your host system.

1. Install Required Dependencies

Install the following files provided by the instructor:

- **VC_redist.x64**
 - Locate `VC_redist.x64.exe` (from the folder shared by the instructor)
 - Right-click → Run as administrator

- **Python 3.12.1 (64-bit)**

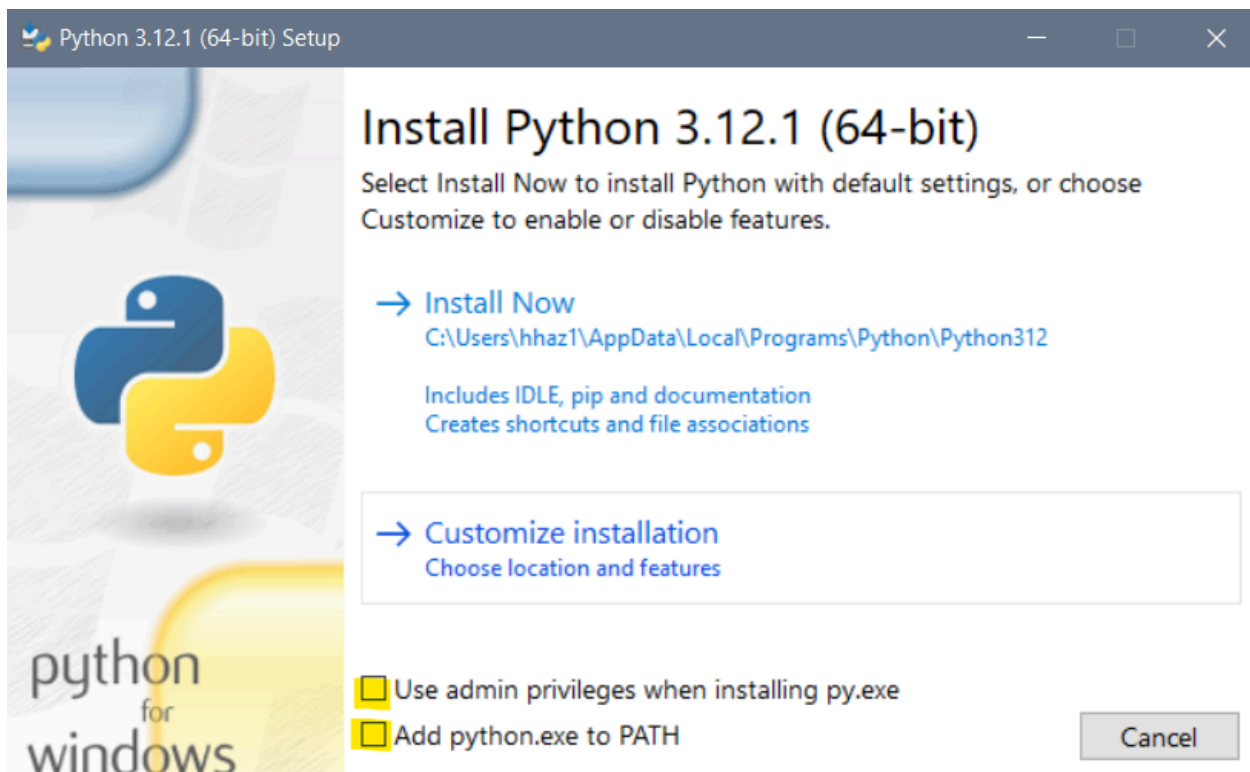
link:

<https://www.python.org/downloads/release/python-3121/>

choose windows installer:

During installation:

- Enable **"Use admin privileges when installing py.exe "**
- Enable **"Add commands directory to PATH**
- At the end, click **"Disable path length limit"**



2. Install Required Python Package

Open **Command Prompt (as Administrator)** and run:

```
pip install pywin32
```

Verification

Confirm Python is installed correctly:

```
python --version
```

confirm **VC_redist.x64** is installed:

1. Open **Settings** → **Apps** → **Installed apps**
2. Search for: Microsoft Visual C++ 2015-2022 Redistributable

If it exists → done.

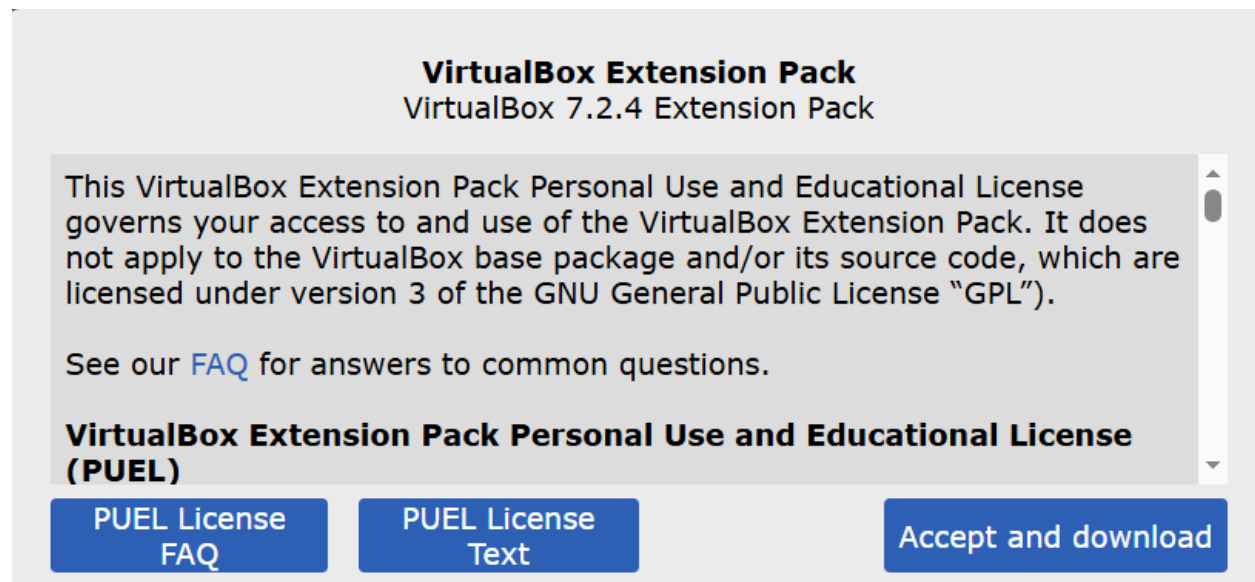
Phase 2: Install VirtualBox

3. Install VirtualBox

- Install **VirtualBox 7.2.4 (Windows, AMD64)**
- Download from: <https://www.virtualbox.org/wiki/Downloads>

4. Install VirtualBox Extension Pack

- Install **Oracle_VM_VirtualBox_Extension_Pack-7.2.4** from the same link



Phase 3: Prepare Virtual Machine Files

5. Extract VM Files

- Extract the provided **Metasploitable** folder
- Extract the provided **Kali Linux** folder

Ensure the files are fully extracted before continuing.

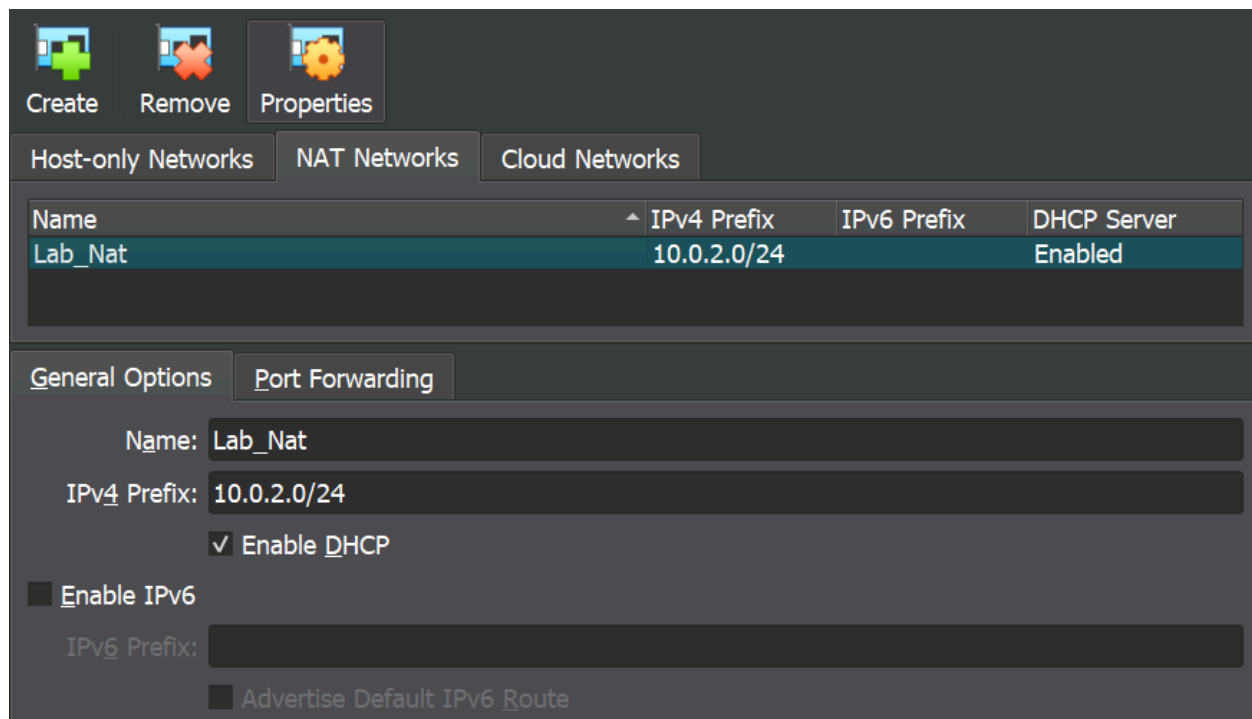
Phase 4: Create a Virtual Network

6. Create a NAT Network

A NAT Network allows virtual machines to communicate with each other **and** access the internet.

Steps:

1. Open **VirtualBox**
2. Go to **File → Tools → Network Manager**
3. Select the **NAT Networks** tab
4. Click **Create**
5. (Optional) Rename the network to `Lab_NAT`



Phase 5: Set Up Metasploitable VM

7. Create the Metasploitable Virtual Machine

1. Click **New**
2. Configure:
 - **Name:** Metasploitable
 - **Type:** Linux
 - **Version:** Ubuntu (64-bit)
3. Allocate resources:
 - **RAM:** Minimum 2 GB (4 GB recommended)
 - **CPU:** At least 1 core
4. from the hard disk section: select **Use an existing virtual hard disk file**
5. Click **Add**, choose the extracted `.vmdk` file, then **Choose**
6. Click **Finish**

8. Configure Network

- Go to **Settings** → **Network**
 - Set **Attached to:** NAT Network
-

Phase 6: Set Up Kali Linux VM

9. Import Kali Linux

1. Click **Add**
2. Select the Kali Linux virtual machine file
3. Import the VM

10. Configure Network

- Go to **Settings** → **Network**
 - Set **Attached to:** NAT Network
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Phase 7: Start & Validate the Environment

11. Start Both Virtual Machines

- Start **Metasploitable**
- Start **Kali Linux**

12. Verification Checklist

Inside **Kali Linux**, verify:

- Both VMs are running
- Kali has an IP address
- Kali can reach Metasploitable
- Kali has internet access

Run:

```
ip a  
ping <Metasploitable_IP>  
ping 8.8.8.8
```

Common Issues & Fixes

Low or Stretched Kali Resolution

Inside Kali:

- Open **Top-left menu** → **Settings** → **Settings Manager** → **Display**
- Adjust the resolution

Clipboard (Copy/Paste) Not Working

If clipboard sharing fails:

1. Power off the vm
2. Go to **Devices** → **Shared Folders** → **Shared Folder Settings**
3. Add a shared folder
4. Use this folder to transfer files between host and Kali.

Phase 8: Prepare SEED Lab Environment (SQL Injection)

In this lab environment, we will also use SEED Lab to demonstrate and practice SQL Injection attacks. SEED Lab provides intentionally vulnerable web applications designed for hands-on security education.

13. Download SEED Lab Virtual Machine

For this lab, download the **SEED Lab virtual machine** from the following link:

SEED Lab VM:

<https://drive.google.com/file/d/12l8OO3PXHjUsf9vfjkAf7-l6bsixvMUa/view>

14. Create the SEED Lab Virtual Machine

1. Open **VirtualBox** and click **New**.
 2. Configure the VM:
 - Name: SEED
 - Type: Linux
 - Version: Ubuntu (32-bit)
 3. from the hard disk section: select **Use an existing virtual hard disk file**.
 4. Click **Add** at the top and select **SEEDUbuntu-16.04-32bit.vmdk** (do not select the **s00X** files).
 5. Click **Finish**
- Then you are ALL SET!

